Lab 2: Introduction to ARM assembly language with Data Segment

Goals

4. Understanding of the fundamental concepts of Assembly Language
5. Able to write Simple Assembly program using ARM v7 ISA
6. Develop familiarity with a Computer System Simulator

Summary of Tasks:

5. Consult ARM Instruction Set Manual available at Canvas
6. Understand the following assembly example
    a. The meaning of each of the identifiers in the program
    b. How data segments and instruction segments work
7. Implement the following two program:
    a. Write a program that will will find the both maximum of 20 integer numbers, show the result in a specific register (e.g. r8)
    b. Write another program that will find the minimum of the same 20 integer numbers and show the results in specific registers (e.g. r9)
8. Do Something cool. The following are a few examples.
    a. Generate the numbers using for loop rather than as a data segment
    b. Display the results in 7-segment display

Assembly program example:

```
.org 1000 @program start at memory address 1000
.data
numbers:
        .word 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24,
25, 26, 27, 28, 29, 30
num_count:
        .word 30

   .text
         .global _start

_start:
   @ Initialize the sum to 0
   mov r0, #0

   @ Load the number of elements (30) into r1
   ldr r1, =num_count
   ldr r1, [r1]
```

```
    @ Load the address of the numbers array into r2
    ldr r2, =numbers

loop:
    @ Load the current number into r3
         ldr r3, [r2], #4  @ Load the value at the current address and increment the
@address by 4 bytes

    @ Add the current number to the sum in r0
        add r0, r0, r3

    @ Decrement the counter (r1) by 1
         subs r1, r1, #1

    @ Check if the counter has reached 0 (end of loop)
        bne loop  @ Branch to 'loop' if r1 is not equal to 0

    @ The loop is done; r0 contains the sum

    @ Put your further code here or return from the function if needed
    @ For example, to exit a program in ARM assembly, you can use the following:
        mov r7, #1  @ syscall number for "exit"
        swi 0       @ initiate the syscall

    .end
```

Lab Demo

Lab 2: Introduction to ARM assembly language

Student Names: ____________________________________

Student IDs: ____

Time & Date: ____________________________

Demo Part 3, Part 4 (optional) to lab instructor. And answer the following questions

5. Do you know that ARM V7 does not have a divide instruction? What could be the possible reason? (Hint. Divide instruction is not very common)

6. Write a pseudo algorithm that can perform divide instruction with other operations.

7. How can we display output using a 7 segment display?